



Solving Trigonometric Equations

You can use inverse functions to solve trigonometric equations that arise in real-world applications.

Solving a Trigonometric Equation Analytically

How to Use a Trigonometric Equation to Solve a Real-World Problem

- Step 1** Sketch the scenario, labeling the given and unknown values.
- Step 2** Write an equation using a trigonometric function that involves the unknown value.
- Step 3** Use an inverse trigonometric function to solve for the unknown value.

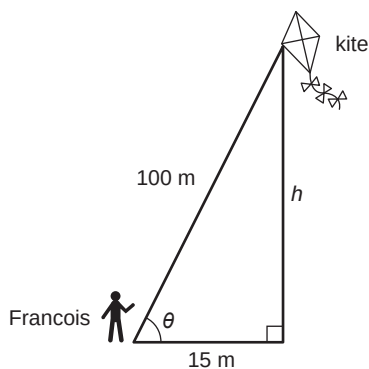
Using an Inverse Trig Function to Solve an Application Problem

EXAMPLE

Francois is flying a kite. He has let out all 100 m of his kite string, and the kite is directly above a spot that is 15 m away from where Francois is standing. What is the angle of elevation of the kite, to the nearest degree, from his location?

SOLUTION

Step 1 Sketch the scenario, labeling the unknown angle as θ .



Step 2 The known side lengths are for the hypotenuse and the leg adjacent to θ , so write an equation in terms of $\cos \theta$.

$$\cos \theta = \frac{15}{100}$$

Step 3 Solve the equation using arccosine.

$$\begin{aligned}\cos \theta &= \frac{15}{100} \\ \theta &= \arccos\left(\frac{15}{100}\right) \\ \theta &\approx 81.37^\circ\end{aligned}$$

The angle of elevation of the kite is about 81° . ■